Antonio Aciobanitei

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Research interests: Probabilistic Machine Learning, Explainable AI, Graph-structured Learning

EDUCATION

University of Westminster **London, UK**

Bachelor of Science in Computer Science (First Class Honours) *September 2022 - July 2025*

- **Relevant Coursework:** Machine Learning and Data Mining, Algorithms: Theory, Design and Implementation, Applied AI, Operational Research and Optimisation
- **Invited Panelist:** Selected to represent the Department of Computer Science at the University Open Days (October-November 2025), presenting on student research and academic pathways.

The Hong Kong Polytechnic University **Hong Kong**

AI Summer School - Grade (A) *July 2024 - August 2024*

- Achieved grade A (96%) and presented applied AI solutions for medical use cases.

RESEARCH EXPERIENCE

University of Cambridge (Accelerate Science Team) **Cambridge, UK**

Google DeepMind Research Intern *July 2025 - August 2025*

- Engineered a large-scale graph of global science to model and interpret the latent structure of research fields.
- Processed a ≈330 million co-citation network with a disparity-filter backbone ($\alpha = 0.05$), yielding ≈170,000 journals and ≈30 million statistically significant links.
- Developed a machine learning embedding pipeline (Node2Vec to UMAP or t-SNE) as the baselayer, and added an overlay of ≈500 Cambridge programme participants (≈35,000 publications) via publication-centroid positions for interpretable exploration of disciplinary patterns.

Recognition: Among the highest-voted departmental Showcases; invited for display at the Royal Academy of Engineering;

Independent Research - Mechanistic Interpretability December 2025

- Conducted an exploratory mechanistic interpretability study on task conditioning in Llama-3-8B-Instruct using attention analysis and targeted head ablation.
- Observed disjoint sufficient attention-head subsets supporting instruction following under task reformulation, validated with random-head and nearby-layer baselines.

University of Westminster **London, UK**

Final-Year Project - Process Mining for Aviation Simulation *September 2024 - April 2025*

- Extended a multimodal transport simulation platform for the aviation sector with process-mining capabilities, enabling event log-based workflow analysis and anomaly detection.
- Invited to present the work at the university's 8th Annual Student-Staff Research Conference (2025).

PUBLICATIONS & MANUSCRIPTS

- From Simulation to Operational Insight: Process Mining for Traceability and Efficiency in Agent-Based ATM Models - manuscript under review (extended version of final-year project)

HONOURS & AWARDS

Google DeepMind Scholar (Martingale Foundation) Awarded for 2026

- Secured full funding (tuition + stipend) for postgraduate study.
- Selected for academic excellence and research potential in Artificial Intelligence.

Hackathons 2025 - 2026

- **Winner / Finalist / Honourable Mention:** Recognised across competitive hackathons (Hack the Future; Iterate London RL; xAI Grokathon) for technical merit in ML systems.
- Built ML systems spanning robust RL control, graph-based simulation, and representation clustering.

Undergraduate of the Year Awards (UK) 2023

- Finalist: Ranked in the top 10 UK undergraduates; recognised for leadership and technical achievement.

SELECTED PROJECTS

Bayesian Weather Forecasting *January 2025 - April 2025*

- Developed Bayesian probabilistic forecasters (heteroscedastic BNNs and Deep Ensembles) for next-day maximum temperature, achieving 93% coverage on 95% prediction intervals with reliable calibration.
- Outperformed deterministic baselines (LightGBM, ARIMA) with 20-25% lower RMSE (2.3C vs 2.9-3.0C) and 15% lower CRPS (1.65 vs 1.95).
- Engineered a reproducible pipeline with automated NOAA/GHCN ingestion, feature engineering (lags, rolling stats, calendar effects), and YAML configs.
- Applied probabilistic training best practices, including CRPS-based early stopping and automated calibration diagnostics (reliability diagrams).

Reinforcement Learning for xArm6 Robotic Manipulation *September 2023 - October 2023*

- Constructed a custom reinforcement learning framework using PyTorch to train the xArm6 robotic arm, achieving a 99% success rate in object manipulation tasks after training on 10,000+ episodes in a simulated environment.
- Developed a framework to compare different training methods (e.g., deep Q-learning and policy-gradient) algorithms, benchmarking key performance metrics to guide optimal strategy selection.
- Utilised PyBullet for real-time physics simulation, enabling dynamic task learning and improving simulation-to-reality transfer for robotic manipulation tasks.

LEADERSHIP & OUTREACH

University of Westminster & Technology Society **London, UK**

Vice President *September 2023 - July 2025*

- Directed 12 technical workshops on AI/ML, algorithm design, and project management, expanded society membership to 50+ in its first semester.
- Mentored society members in software development, algorithms, and collaborative coding projects, fostering a culture of technical excellence and innovation.
- Led onboarding initiatives for 36 new computer science students, maintaining full retention across the first semester.

TECHNICAL SKILLS

ML & Data: Python, scikit-learn, PyTorch, UMAP, t-SNE, Node2Vec (PecanPy), Pandas, NumPy, BNNs, Gaussian Processes, Variational Inference

Infrastructure & Visualisation: Docker, AWS, Linux, Git, GitHub Actions, DuckDB, Plotly, React, Next.js